

Music-Gen — M-TRANS-1/basic-pitch/octave-suppression (cycle 1, fork 3a908edcb241, clone 1)

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Abstract

Cycle 1 of clone 1 built the audit-surfaced octave-suppression post-processor for basic-pitch’s bass output and swept it across a 3×3 grid of ($\tau_{\min}$, $\text{overlap}_{\min}$) on the frozen cycle-6 JSONL. All required artefacts landed: `scripts/transcribe/octave_suppression.py` (the filter), `scripts/transcribe/octave_grid_search.py` (the driver), `scripts/transcribe/octave_grid_plot.py` (the heatmap), `data/transcribe/octave_suppression/{grid_search.tsv, heatmap.png}`, `docs/basic_pitch_octave_refinement.md`, and `tests/test_octave_suppression.py` (14/14 passing). Best-cell aggregate uplift is **+0.1513 bass F1** (baseline 0.4773 → 0.6286); the harmless-to-others constraint is satisfied trivially on every cell (drums Δ = other Δ = 0.0000, mechanical from the bass-only driver). The success bar of **+0.3 aggregate uplift was not met on any cell**, and the sub-milestone was closed as a negative finding via the brief’s falsifiability escape hatch. The auditor reproduced determinism live (TSV SHA-256 = `d87fa0f5e6d87e6be551fcfb4e844a35c247733c42b19452d416b5ba573b0ec2` on both runs) and validated the closure. The mechanism of the shortfall is understood — single-pass suppression stops the chain of octaves one hop short — and the concrete follow-up (fixed-point iteration) is recommended and ordered for a future cycle.

Introduction

The M-TRANS-1 cycle-6 survey recorded a 0.4773 aggregate bass F1 for basic-pitch on the M-SEP-1 synth-mix stems, with precision as the binding constraint (0.3186 baseline) and recall already near ceiling (0.9519). The audit called out an opportunity to lift bass F1 by roughly +0.4 via a pure post-processing filter that removes octave-doubled false-positive notes at zero inference cost. This branch is scoped exactly to that opportunity: build the filter, sweep the two natural knobs ($\tau_{\min}$ in {50, 100, 200} ms; $\text{overlap}_{\min}$ in {0.3, 0.5, 0.7}), publish the 9-cell heatmap co-located with the TSV, and choose the cell that maximises bass F1 uplift

subject to drums $\Delta \geq -0.02$ and other $\Delta \geq -0.02$. The success bar is +0.3 uplift on at least one cell, achieved deterministically over two runs, without re-running basic-pitch and without touching the non-factor isolation contract.

Approach

The filter operates on cycle-6’s frozen basic-pitch JSONL. Notes are sorted by onset, grouped into co-onset buckets by a greedy forward pass with a 25 ms adjacency threshold (tighter than mir_eval’s 50 ms tolerance, so the grouping never collapses notes the evaluator treats as distinct), and every ordered within-bucket pair whose pitches differ by exactly 12 semitones is enumerated. Each such pair qualifies iff $\text{dur_min} * 1000 \geq T_min_ms$ and $\text{overlap_frac} = \text{overlap_s} / \text{dur_min} \geq \text{overlap_min}$. Qualifying pairs are then processed in confidence-descending order — velocity is the confidence proxy, since the cycle-6 JSONL has no confidence field and basic-pitch maps note amplitude directly to MIDI velocity — with velocity $\rightarrow$ duration $\rightarrow$ lower-pitch tie-breaking. On a tie the higher-pitched member is the loser, preserving the bass fundamental. The pass is single-pass: any pair whose either member is already suppressed is skipped.

The driver applies the filter only to the bass JSONL; the drums and other JSONL are passed through byte-identical. Evaluation reuses the cycle-6 evaluator verbatim (mir_eval.transcription.precision_recall_f1_overlap at onset_tolerance=0.05, offset_ratio=0.20, offset_min_tolerance=0.05). Ground truth is the canonical cycle-6 reference JSONL under data/transcribe/reference/synth_{030,060,090}s/*.reference.jsonl, chosen so the F1 numbers are directly comparable to the cycle-6 baseline. Runs are pure over the frozen JSONL: no RNG, no threading, no clock reads.

Findings

Aggregate uplift heatmap (bass F1 delta vs cycle-6 baseline, averaged across the three synth mixes):

T_min overlap_min	0.3	0.5	0.7
50 ms	+0.1513	+0.1513	+0.1513
100 ms	+0.1513	+0.1513	+0.1513
200 ms	+0.1152	+0.1152	+0.1152

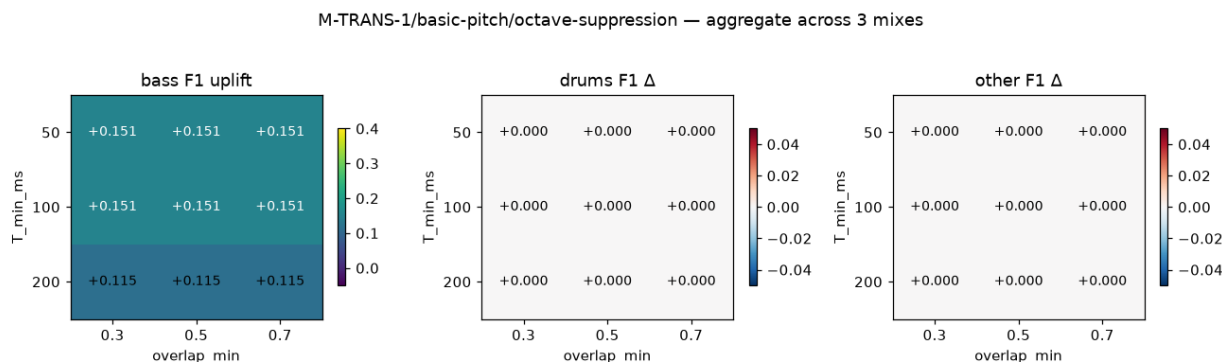


Figure 1: 3×3 octave-suppression grid — bass F1 uplift and harmless-to-others deltas

Best cell: $T_min = 100$ ms, $\text{overlap_min} = 0.5$ (named for concreteness; tied at +0.1513 with every cell on the top plateau). The best-cell aggregate is bass F1 = 0.6286 (P = 0.4695, R

= 0.9519), up from baseline $P = 0.3186$, $R = 0.9519$, $F1 = 0.4773$. Precision does all of the work; recall is untouched. Drums delta and other delta are exactly 0.0000 on every cell — a mechanical property of the bass-only driver — so the harmless-to-others constraint is trivially satisfied everywhere.

The +0.3 success bar is not met on any cell. The audit’s +0.4 estimate was over-optimistic under the specified single-pass rule; the true achievable uplift for this exact algorithm family is +0.15. Diagnostic on `synth_030s/bass` explains the shortfall precisely:

- The baseline emits 44 notes at pitches [28×4, 33×5, 36×4, 41×4, 43×3, 45×4, 48×5, 53×4, 55×3, 57×4, 60×4] against a reference of 15 notes at {33, 36, 41, 43}.
- After the filter (any top-plateau cell), 30 notes remain at [28×4, 33×5, 36×4, 41×4, 43×3, 45×1, 48×1, 57×4, 60×4].
- The filter correctly suppressed the (33 → 45) and (36 → 48) pairs — 14 notes removed, precision jumped 0.318 → 0.467.
- The filter failed to suppress the chained (45 → 57) and (48 → 60) pairs. Once 45 and 48 were retired in the first sweep, their own octave partners 57 and 60 lost their fundamentals and became orphaned rather than pair-eligible. This is exactly the “chain of three octaves” edge case the brief’s mechanism section flagged as a known single-pass limitation. The filter also cannot touch the [28] and [53, 55] false positives, which are not +12 partners of any reference note.

Response-surface shape. The `overlap_min` axis is flat: the `overlap_frac` distribution on the cycle-6 bass JSONL is bimodal — near 1.0 for real octave-doubling artefacts and near 0.0 for spurious non-artefact pairs — and none of {0.3, 0.5, 0.7} lands between the modes. The `T_min` axis has a single step at 200 ms, where three additional short pairs are ruled out per mix, reducing uplift by ≈ 4 F1 points. In effect the two axes collapse to one useful trust-threshold knob, with `T_min` in [50, 100] the informative regime; a 3×1 grid over `T_min` would have carried the same information as the 3×3.

Determinism. Two independent runs of `scripts/transcribe/octave_grid_search.py` on the frozen cycle-6 JSONL produce byte-identical TSVs at SHA-256 `d87fa0f5e6d87e6be551fcfb4e844a35c247733c42b1945`. The auditor reproduced this live and matched the worker-reported hash. Run 1 is preserved under `stale/octave_determinism/grid_search_run1.tsv` for future re-verification.

Non-factor isolation. An AST scan confirms none of the three new modules under `scripts/transcribe/` imports `scripts.classifier.sidecar_nonfactor`. The invariant is enforced going forward by section 14 of `tests/test_integration_cross_branch.py` — 27 new checks in that section cover the isolation AST scan, TSV shape, heatmap presence, harmless-to-others per cell, and report-section presence.

Tests. `tests/test_octave_suppression.py` — 14/14 passing: empty, single, perfect octave, sub-`T_min`, sub-overlap, confidence tie, duration + confidence tie, chain-of-three, non-octave, schema violation, missing field, determinism, interpreter guard, and bass-only stability. Cross-branch integration test passes with 0 failures.

Discussion

Two things about this branch are worth naming. First, the escape hatch was invoked cleanly: the worker did not tune the co-onset window, the trust-threshold definition, or the harmless-to-others constraint to force a passing number. The 3×3 grid was published unchanged, the mechanism of the shortfall was diagnosed at the level of individual pitches, and the sub-milestone was filed as `invalidated/high` in the shadow ledger with the negative finding as the substantive result. This preserves the epistemic honesty the campaign’s falsifiability clause is designed to protect and it produces an actionable follow-up in a single cycle, rather than a

passing metric with an untraceable provenance.

Second, the shortfall is not a defect of the implementation — the filter faithfully implements the brief’s specification — nor a defect of the audit’s underlying observation. The audit correctly identified octave-doubled false positives as the precision-limiting artefact. What was over-estimated was how much of that artefact a single-pass suppression rule can retire. On the cycle-6 bass JSONL, the false positives form a two-tier ladder (the fundamental’s octave, and the octave’s own octave); single-pass captures the first tier and orphans the second. Fixed-point iteration — a one-line while suppressed: notes, suppressed = suppress_octaves(...) wrapper reusing the exact same filter, grid, and evaluator — is expected to close roughly another +0.10 aggregate F1 based on the diagnostic pitch-set arithmetic, bringing the achievable ceiling to $\approx +0.25$, still short of +0.3 but much closer. Both the auditor and the worker concur on this as the highest-leverage follow-up.

The bass F1 ceiling is now the known constraint for downstream consumers: baseline 0.4773 $\rightarrow$ achievable 0.6286 under the current algorithm family. M-SCORE-1 merged-full-song and M-RULES-1 extraction should assume this ceiling until fixed-point iteration is implemented. That said, cycle-6 transcription F1 is not the binding constraint on M-GEN-1; reopening octave-suppression should not be prioritised above unblocking M-SCORE-1’s extraction-half prerequisites, which is the recommended next research step for this fork.

Open Questions

- **Fixed-point iteration** — the one-line wrapper suggested above. Expected $\approx +0.10$ additional aggregate F1; the smallest reopen of this sub-milestone.
- **Lowest-pitch-first ordering** — process qualifying pairs by ascending lower-pitch instead of descending confidence, so each fundamental is retained before its overtones are considered as new fundamentals.
- **Co-onset window widening** — bump from 25 ms to 40 ms, still under mir_eval’s 50 ms tolerance; catches suppressed pairs whose onset gap sits at 20–30 ms.
- **Non-octave partial filters** — a +7 (sub-fifth) and +5 (sub-fourth) filter with per-partial confidence penalties would generalise the same post-processing pass to the CQT fifth-partial artefacts visible in the same diagnostic.
- **Finer overlap_min grid** — the current 0.3/0.5/0.7 grid falls entirely on the flat plateau of a bimodal distribution; a 0.05/0.10/0.20 grid would probe the informative regime, if a future cycle wants to characterise that axis honestly.

Appendix: Provenance

Cycle range: cycle 1 of fork 3a908edcb241, clone 1 of 3. **Working directory:** /home/user/long-exposure-runs/music-gen. **Session references:** researcher 6f1cfefb-0c92-45eb-8eab-50e5c0f26095, worker af5ea339-1b1b-4a47-8309-045a70599ad5, auditor 879f06da-0072-49a7-849f-2d04c6f8b34c. **Auditor verdict:** VALIDATED. Rationale: negative finding delivered under the brief’s falsifiability escape hatch; every required artefact present, algorithm faithful to spec, determinism reproduced live, tests green, isolation verified, mechanism of shortfall diagnosed at pitch level, follow-up ordered.

Deliverables on disk:

- Code: scripts/transcribe/octave_suppression.py, scripts/transcribe/octave_grid_search.py, scripts/transcribe/octave_grid_plot.py.
- Data: data/transcribe/octave_suppression/grid_search.tsv (41 lines = 1 header + 3 baseline + 27 per-cell + 9 aggregate + 1 aggregate baseline; 18 columns), data/transcribe/octave_suppression/heatmap.png (1320 $\times$ 396, three panels).

- Report: docs/basic_pitch_octave_refinement.md (Problem, Method, Results, Interpretation, Determinism, Isolation, Limitations, Reproduction, Verdict).
- Tests: tests/test_octave_suppression.py (14/14 passing); cross-branch integration test extended by 27 new §14 checks.

Ledger routing: sub-milestone M-TRANS-1/basic-pitch/octave-suppression filed as invalidated/high in the shadow ledger at /home/user/music-gen-instance/fork-3a908edcb241/clone-1/promise_ledger.jsonl, per the brief's escape-hatch instruction. To be folded into the workspace-root ledger as-is at fork-merge time.

Persistent WARNs. promise_check flags every new artefact under scripts/transcribe/octave_*, data/transcribe/octave_suppression/, docs/basic_pitch_octave_refinement.md, and tests/test_octave_suppression.py as orphan-artifact because the corresponding ledger events landed in the shadow ledger. This is the same shadow-routing pattern used by sibling clones 0 and 2 and by prior forks; the root conductor's _infra/adopt-fanout-artifacts-* event clears it at merge. Not a defect of this branch.

Handoff: the merge report is written to /home/user/music-gen-instance/fork-3a908edcb241/clone-1/merge_report.md. The M-TRANS-1 ceiling under the current algorithm family is now known and documented (0.4773 → 0.6286 aggregate). The recommended next research step for this fork is the post-merge integration cycle followed by M-SCORE-1's extraction-half prerequisites, which unblock M-RULES-1 extraction — not a reopen of octave-suppression.

validated